

# Nonlinear Solid Mechanics

A COMPILATION OF  
NOTES, NOTATIONS, AND  
ILLUSTRATIONS

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# Abstract

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# Acknowledgements

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# Contents

|                                                      |             |
|------------------------------------------------------|-------------|
| <b>Abstract</b>                                      | <b>i</b>    |
| <b>Acknowledgements</b>                              | <b>ii</b>   |
| <b>List of Tables</b>                                | <b>vi</b>   |
| <b>List of Figures</b>                               | <b>viii</b> |
| 0.1 Shock . . . . .                                  | 1           |
| 0.1.1 Introduction . . . . .                         | 1           |
| 0.1.2 An Ideal Fluid . . . . .                       | 1           |
| 0.1.3 Definition of a Shock . . . . .                | 1           |
| 0.1.4 Conservation Laws for a Single Shock . . . . . | 1           |
| 0.1.5 $PV$ Diagram . . . . .                         | 5           |
| 0.1.6 $PU_P$ Diagram . . . . .                       | 6           |



# List of Tables



# List of Figures

|   |                                                                                                                           |   |
|---|---------------------------------------------------------------------------------------------------------------------------|---|
| 1 | Piston moving left-to-right along the $x$ -axis, with piston speed $U_P$ , causing shock with shock speed $U_S$ . . . . . | 1 |
| 2 | Pressure as a function of volume for the shock Hugoniot locus of thermodynamic equilibrium states. . . . .                | 5 |

## 0.1 Shock

### 0.1.1 Introduction

### 0.1.2 An Ideal Fluid

### 0.1.3 Definition of a Shock

### 0.1.4 Conservation Laws for a Single Shock

#### Laboratory Fixed Coordinate Frame

The derivation that follows assumes a **laboratory-fixed reference frame**, as shown in Fig. 1.<sup>1</sup> Consider a rigid tube containing a rigid piston and an ideal fluid at ambient density  $\rho_0$  and ambient pressure  $P_0$  at time  $t = t_0$ .

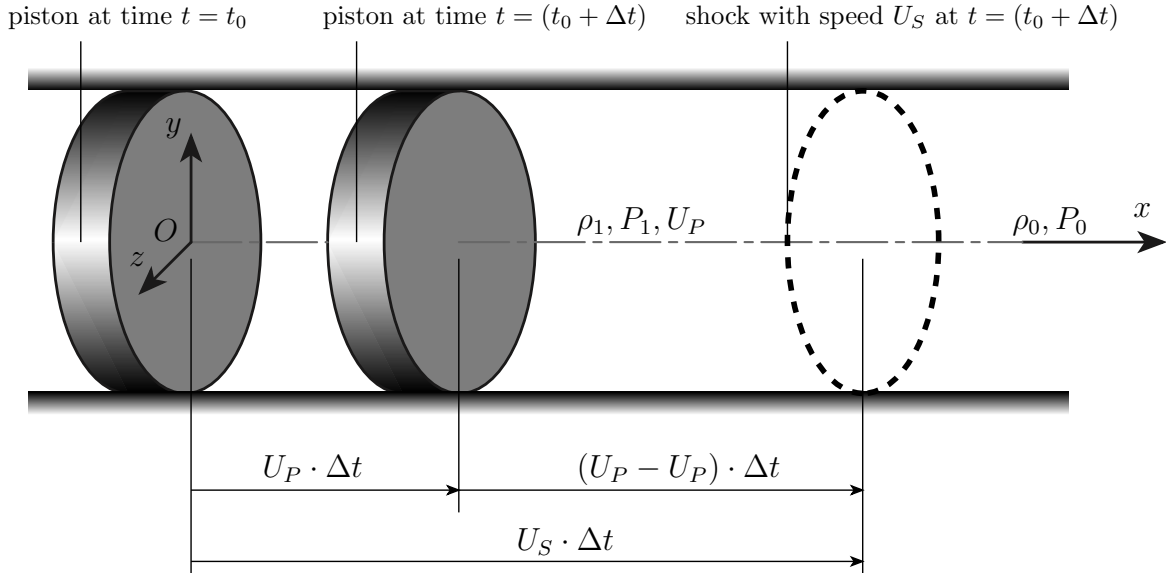


Figure 1: Piston moving left-to-right along the  $x$ -axis, with piston speed  $U_P$ , causing shock with shock speed  $U_S$ .

Assume at time  $t = (t_0 + \Delta t)$ , the piston moves left-to-right along the  $x$ -axis with speed  $U_P$ , measured in the fixed laboratory reference frame. Assume the piston speed  $U_P$  causes a shock to propagate through the material with speed  $U_S$ . We require the following:

$$0 < U_P < +\infty \quad (1)$$

$$0 < U_S < +\infty \quad (2)$$

$$U_P < U_S \quad (3)$$

<sup>1</sup>A shock-fixed reference frame derivation is developed in a subsequent section.

From these considerations, we conclude that the material to the left of the shock wave in Fig. 1 moves to the right at speed  $U_S$ . The material to the right of the shock wave remains at rest.<sup>2</sup>

Since the piston compresses the material in the tube, we know

- $\rho_1 > \rho_0$ ,

where  $\rho_1$  is the density of the material to the left of the shock.

Allow some time interval  $\Delta t$  to elapse, as illustrated in Fig. 1. Consider the distances along the  $x$ -axis, written as a multiplication of speeds and elapsed time. In the time interval  $\Delta t$ , the piston moves along the  $x$ -axis a distance of  $U_P \cdot \Delta t$ . In the same time interval, the shock wave moves along the  $x$ -axis a distance of  $U_S \cdot \Delta t$ . Thus, the control volume at  $t = (t_0 + \Delta t)$  has an axial length of  $(U_S - U_P) \cdot \Delta t$ .

From these axial lengths and piston cross-sectional area  $A$ , we can write the following relationships:

- volume

$$V_0 = V|_{t=t_0} = A \cdot U_S \cdot \Delta t,$$

$$V_1 = V|_{t=(t_0+\Delta t)} = A \cdot (U_S - U_P) \cdot \Delta t,$$

- density

$$\rho_0 = \rho|_{t=t_0},$$

$$\rho_1 = \rho|_{t=(t_0+\Delta t)},$$

- mass

$$\rho_0 V_0 = \rho_0 \cdot A \cdot U_S \cdot \Delta t,$$

$$\rho_1 V_1 = \rho_1 \cdot A \cdot (U_S - U_P) \cdot \Delta t.$$

### Conservation of Mass

Conservation of mass within the control volume requires that mass at time  $t = (t_0 + \Delta t)$  be unchanged from mass at time  $t = t_0$ . Thus

$$\rho_0 \cdot A \cdot U_S \cdot \Delta t = \rho_1 \cdot A \cdot (U_S - U_P) \cdot \Delta t, \quad (4)$$

or

$$\boxed{\rho_0 U_S = \rho_1 (U_S - U_P), \quad \text{(Conservation of Mass)}}. \quad (5)$$

---

<sup>2</sup>Furthermore, we assume the material throughout the tube and for all time is inviscid, non-diffusive, and non-reacting. We assume an adiabatic boundary, thus no heat conduction occurs through the boundary of the tube walls or piston. We assume no heat transfer from radiation occurs across the shock. Finally, we assume body forces from gravity and electromagnetism are negligible.

Define the **specific volume** as

$$V \triangleq \frac{1}{\rho}. \quad (6)$$

Then, conservation of mass be rewritten as

$$V_1 = V_0 \left( 1 - \frac{U_P}{U_S} \right), \quad \text{(Conservation of Mass, Volume Form).} \quad (7)$$

From (1)–(3), we know  $V_1 < V_0$ , which makes sense since we are compressing the material and generally have the understanding that increased pressures causes decreased volumes.

### Conservation of Momentum

The net driving force from the piston on the material is  $(P_1 - P_0) \cdot A$ . Conservation of momentum requires  $F = d(ma)/dt$ . Written over the time interval  $\Delta t$  and for constant mass, conservation of momentum can be written

$$F \cdot \Delta t = m \cdot \Delta U. \quad (8)$$

Rewriting this expression in terms of pressures and densities, we have

$$(P_1 - P_0) \cdot A \cdot \Delta t = (\rho_0 \cdot A \cdot U_S \cdot \Delta t) (U_P - 0) \quad (9)$$

which gives

$$P_1 - P_0 = \rho_0 U_S U_P, \quad \text{(Conservation of Momentum).} \quad (10)$$

Using (6), (10) can be rewritten as

$$P_1 - P_0 = \frac{U_S U_P}{V_0}, \quad \text{(Conservation of Momentum, Volume Form).} \quad (11)$$

### Conservation of Energy

The work done on the control volume by the piston moving left-to-right is given by

$$W_P = \underbrace{P_1 \cdot A}_{\text{force}} \cdot \underbrace{U_P \cdot \Delta t}_{\text{disp.}}. \quad (12)$$

Note that the right boundary of the control volume, located at the shock location, does not move, so the work across this boundary is identically zero.

The kinetic energy increase of the control volume is simply the one-half times the mass times the square of the change in velocity, thus

$$KE = \frac{1}{2} \underbrace{(\rho_0 \cdot A \cdot U_S \cdot \Delta t)}_{\text{mass}} \underbrace{(U_P - 0)^2}_{\text{velocity}}. \quad (13)$$

Finally, the increase in internal energy is just the internal energy **per unit mass**,  $e$ , multiplied by the mass,

$$IE = (e_1 - e_0) \underbrace{(\rho_0 \cdot A \cdot U_S \cdot \Delta t)}_{\text{mass}}. \quad (14)$$

Equating these quantities gives

$$P_1 \cdot A \cdot U_P \cdot \Delta t = \frac{1}{2} \rho_0 \cdot A \cdot U_S \cdot \Delta t \cdot U_P^2 + (e_1 - e_0) \rho_0 \cdot A \cdot U_S \cdot \Delta t \quad (15)$$

or simply

$$\boxed{P_1 U_P = \rho_0 U_S \left( \frac{U_P^2}{2} + e_1 - e_0 \right), \quad \text{(Conservation of Energy)}}. \quad (16)$$

The foregoing equation can be written without kinematic variables  $U_S$  and  $U_P$ . Next, note that conservation of mass can be rewritten as

$$\frac{U_P}{U_S} = 1 - \frac{\rho_0}{\rho_1}. \quad (17)$$

Then, with (16),

$$P_1 U_P \stackrel{(10)}{=} (P_1 - P_0) \frac{U_P}{2} + \rho_0 U_S (e_1 - e_0), \quad (18)$$

$$= -P_0 U_P + 2 \rho_0 U_S (e_1 - e_0), \quad (19)$$

$$(e_1 - e_0) = \frac{U_P (P_1 + P_0)}{2 \rho_0 U_S}, \quad (20)$$

$$\stackrel{(17)}{=} \left( \frac{P_1 + P_0}{2 \rho_0} \right) \left( 1 - \frac{\rho_0}{\rho_1} \right), \quad (21)$$

$$\stackrel{(6)}{=} \left( \frac{P_1 + P_0}{2} \right) V_0 \left( 1 - \frac{V_1}{V_0} \right), \quad (22)$$

which simplifies to

$$\boxed{e_1 - e_0 = \frac{-1}{2} (P_1 + P_0) (V_1 - V_0), \quad \text{(Hugoniot equation)}}. \quad (23)$$

The Hugoniot equation is the Conservation of Energy jump condition with state variables alone; there are no kinematic variables in the equation. To plot the Hugoniot relationship in the  $PV$  plane with  $e_1 > e_0$ , the initial state  $(V_0, P_0)$  must be known.

### Speed-Volume-Pressure Relationships

The alternative forms for mass conservation (7) and momentum conservation (11) readily allow for the shock speed  $U_S$  and particle speed  $U_P$  to be written in terms of volume and pressure. These relationship are found to be

$$U_S = V_0 \sqrt{\frac{P_1 - P_0}{-(V_1 - V_0)}} = V_0 \sqrt{\frac{\Delta P}{-\Delta V}}, \quad (24)$$

and

$$U_P = \sqrt{-(P_1 - P_0) (V_1 - V_0)} = \sqrt{-\Delta P \Delta V}. \quad (25)$$

## Alternative Formulations

Now we develop the shock-fixed reference frame equations for mass, momentum, and energy conservation.

### 0.1.5 $PV$ Diagram

The  $PV$  diagram shows pressure  $P$  as a function of volume  $V$ . Thus, the state variables  $P$  and  $V$  are considered in isolation from the other state and kinematic variables. The initial laboratory state is denoted  $(V_0, P_0)$ . A large compressive pressure  $P_1$  is applied to the material, which causes the volume to shrink to  $V_1$ . If this applied pressure  $P_1$  causes a shock, we record the pressure and volume state, and then repeat the experiment at a different pressure, e.g.,  $P_2$ . Several compression experiments are run, and the  $PV$  states that created a shock in the material are recorded in the  $PV$  diagram.

For concreteness, we show four experiments with applied pressures  $P_1 \dots P_4$  and resultant volumes  $V_1 \dots V_4$ , shown in Fig. 2. Note that all experiments start from the same initial pressure  $P_0$  and initial volume  $V_0$ .

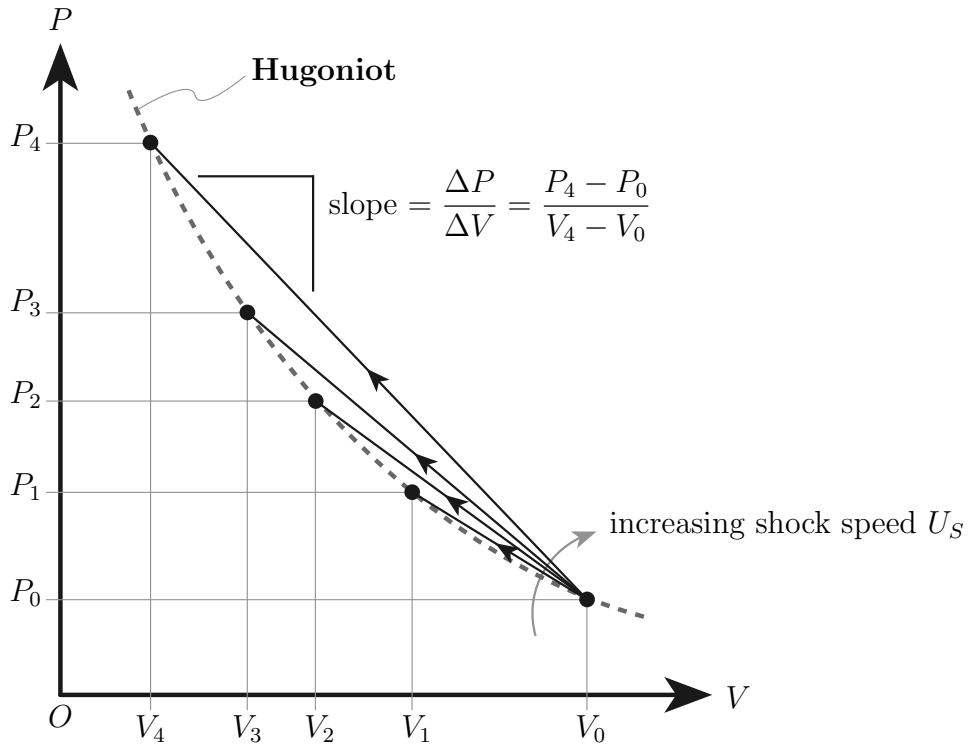


Figure 2: Pressure as a function of volume for the shock Hugoniot locus of thermodynamic equilibrium states.

These four thermodynamic states lie on a locus of all possible thermodynamic states that can exist in the material behind a shock. This locus is referred to as the **shock**

**Hugoniot.**

It is important to emphasize that the locus **is not** the path taken by the material between two thermodynamic states.

Note that (24) describes the mathematical increase in shock speed with increasing slope of the Hugoniot, indicated in Fig. 2. Specifically, the slope of the Hugoniot goes as the negative squared value of the shock speed  $U_S$  per initial volume  $V_0$ , or

$$\frac{\Delta P}{\Delta V} = - \left( \frac{U_S}{V_0} \right)^2. \quad (26)$$

**0.1.6  $PU_P$  Diagram**